Corrected MCQs

1. **D.** The argument must be strictly positive: 2 $x - 1 >$ Zero is excluded.
2. **A.** The logarithm transforms a product into a sum; no analogous rule transforms a product sum of arguments.
3. **C.** The exponential and logarithm functions are converse on their domains.
4. **B.** Apply u'/u with $u = 1 + x^2 > 0$; factor 2 x It's from the compound.
5. **D.** Apply the exponential to both members; $e^2 > 0$ checks the domain.
6. **C.** $2^{-3} = 0{,}125 \geq 0{,}1$ and $2^{-4} = 0{,}0625 < 0{,}1$. The continuation decreases, so no previous rank It's okay.
7. **A.** Take the logarithm then divide by ln $q < 0$ reverses the sense of inequality.
8. **B.** Comparative logarithm growth and positive logarithm power x.
9. **D.** The right member shall impose $x > 0$. For $x \neq 0$, the general identity is $(x^2) = 2\ln|x|$.
10. **C.** $g' = (1 - \ln x)/x^2$, positive before e then negative. The maximum reached in e is equal to $1/e$.

True/false. False: ln 2 $\neq 0$. False: ln 2 > 0. False: maximum 1/ $e < 1$ (question 10) prohibited.

Corrected applications and support

Application 1. Fields: $]-\infty;2[\ ;\ ]-\infty;-1[\cup]1;+\infty[$. For the equation, $x>-1/2$ and $2x+1=1$, where $x=0$.

Application 2. Take $a=b=1$. For $x=-1$, the left limb $\ln 1$ exists but $\ln(-1)$ does not exist not in the real. The good identity is $(x^2)=2\ln|x|$ for $x\neq 0$.

Implementation 3. Area $\mathbb{R}$, derivative $2x/(1+x^2)$, negative before 0 and positive after ; minimum $g(0)=0$, therefore $g\geq 0$. $g(-x)=g(x)$; symmetry confirms the opposite variations on both sides of 0.

Implementation 4. First threshold: $n\geq \ln 2/\ln 1{,}05\simeq 14{,}21$, so 15; $1{,}05^{14}\simeq 1{,}980$ and $1{,}05^{15}\simeq 2{,}079$. Second: $n\geq \ln 0{,}2/\ln 0{,}9\simeq 15{,}28$, therefore 16, $0.9^{15}\simeq 0.2059$ and $0.9^{16}\simeq 0{,}1853$.

Application 5. $\ln(2x)/x=\ln 2/x+\ln x/x\to 0$. $x\ln(3x)=x\ln 3+x\ln x\to 0$.

Implementation 6. $f(x)=\ln x/x$ tend towards $-\infty$ in 0^{+}, grows strictly up to $1/e$, then decreases Strictly around 0 remaining positive after e. Thus: a solution for $a<0$; one for $a=0$ (1) ; two for $0<a<1/e$; one for $a=1/e$ (e) ; none for $a>1/e$. The conclusions make use of the continuity on each branch.

Return on error

For the threshold, ask to first predict if the sequence increases or decreases, then compare
This sense with the inequality obtained. To resume false distributivity with a product of which one
1. The parameter problem allows to work existence, singleness per branch and
limits not reached.

Conduct and correction

Benchmarks for the teacher

S44 is placed before the first white, but the logarithm does not belong to its announced perimeter. Insisting on conditions $a > 0, b > 0$ before grouping logarithms. Differentiability of The derivation formula is shown for a threshold of 0. $< q < 1$, the sign negative ln q Inequality reverses. Control the candidate rank and the precedent.

Corrected activity

1. Each step multiplies the error by 0.7: $e_n = 5(0{,}7)^n$ for $n \geq 0$.

2. $e_n \leq 10^{-4}$ is equivalent to $(0{,}7)^n \leq 2 \times 10^{-5}$. By growth of ln, $n \ln 0{,}7 \leq \ln(2 \times 10^{-5})$. Like $\ln 0{,}7 < 0$, we get

$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0{,}7} \simeq 30{,}33512.$$

   The first suitable integer is 31.

3. $e_{30} \simeq 1{,}1270 \times 10^{-4} > 10^{-4}$ and $e_{31} \simeq 7{,}8888 \times 10^{-5} < 10^{-4}$. The continuation of errors decreases, so this control confirms minimality.

4. For target 10^{-6} the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0{,}7} \simeq 43{,}2465$, so $n = 44$. The actual threshold increased by $\frac{\ln 100}{|\ln 0{,}7|} \simeq 12{,}91$, then round to the next integer. The number of steps is not multiplied by 100.

E1. We need two. $x - 1 > 0$ and $x + 3 > 0$, therefore $x > \frac{1}{2}$. In this field, the focus of ln gives 2 $x - 1 = x + 3$, let $x = 4$. This value belongs to the domain; all solutions are $\{4\}$.

E2. We have to $x > 2$. The formula of the product gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, either $(x-3)(x+1) = 0$. Only $x = 3$ satisfied $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is detrievable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the rule of the product gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, positons $t = \sqrt{x} \to +\infty : \frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, positons $t = -\ln x \to +\infty : x^2 \ln x = -te^{-2t} \to$ These two limits use comparative growth.

E5. For $x > 0$, posisons $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. minimum $g(1) = 0$, therefore $g(x) \geq 0$, which is equivalent to $x \leq x - 1$.

Corrected from DM6

1. In 0^+, the numerator tends to $-\infty$ and denominator to $0^+ : f \to -\infty$. In $+\infty, f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, none in e^{-1} By quotient,

$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$

   So f grows strictly up to 1, then decreases strictly; its maximum is $f(1) = 1$.

2. By drifting $-(\ln x)x^{-2}$, we get $f''(x) = \frac{2 \ln x - 1}{x^3}$. The numerator changes sign, negative to the positive, in $x = \sqrt{e}$. So f is concave on $]0, \sqrt{e}]$, convex $[\sqrt{e}, +\infty[$ and has inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0,1]$, continuity, strict growth, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1,+\infty[$, strict decrease from 1 to 0 gives exactly a second The first one is in $]e^{-1},\frac{1}{2}[$, since $f(e^{-1})=0$ and $f(\frac{1}{2})=2(1-\ln 2)\simeq 0{,}6137>\frac{1}{2}$. The second is in $]5,6[$, since $f(5)\simeq 0{,}5219>\frac{1}{2}>f(6)\simeq 0{,}4653$.

4. The maximum gives $\frac{1+\ln x}{x}\leq 1$. Like $x>0$, you can multiply without changing the meaning: $1+\ln x\leq x$. Equality is the only maximum of f, therefore $x=1$.

Corrected work in the manual or an anal

Logarithm imposes a strictly positive argument.

1. $3x-7>0\iff x>\frac{7}{3}$, therefore $D_f=]\frac{7}{3};+\infty[$.
2. $-x^2+4x-3=-(x-1)(x-3)>0$ exactly when $1<x<3$. Thus $D_g=]1;3[$.
3. It must be simultaneously $x>0$ and $2-x>0$; their intersection gives $D_h=]0;2[$. The coefficient -3 placed in front of the second logarithm does not change its definition condition.